Stresses in External and Internal Unbonded Tendons: Unified Methodology and Design Equations

Zhi-Qi He[1] and Zhao Liu[2]

Abstract: A unified methodology is proposed to compute the tendon stresses in externally prestressed beams in elastic and ultimate states, and practical design equations are developed to predict the ultimate tendon stresses. Previous test results have shown that there is an almost linear relationship between the tendon stress increment and the beam midspan deflection in both elastic and inelastic ranges. The key point is to seek this relationship and then the prediction of the tendon stress increment is transferred to the problem of calculating the beam midspan deflection. The second-order effect of external tendons is included in this deformation compatibility based analysis. Moreover, reasonable simplifications are made to establish practical design equations for the prediction of the ultimate tendon stress. The proposed equations are shown to be able to account for most of the important variables and result in good agreement with test data. It is shown that the ultimate stress increment is mainly dependent on the tendon eccentricity and the neutral axis depth, and the second-order effect can be reasonably taken into account using the stress and depth reduction factors.

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Introduction

In the past decades, external prestressing has been extensively used in the construction of new structures as well as in the strengthening of existing ones because of its attractive advantages. In the design of externally prestressed beams, it is necessary to evaluate the stress in external tendons in order to check the stress status and the deflection of the beam under service limit state and to calculate the nominal flexural resistance of the beam under strength limit state.

The procedure for determining the stresses in external or internal unbonded tendons is different from that in bonded tendons. The main differences lie in two aspects. First, strain incompatibility exists between unbonded tendons and the surrounding concrete for lacking of bond between them. Second, eccentricity variation of external tendons occurs between anchorages and/or deviators for lacking of restraint along section depth, known as the second-order effect that distinguishes the behavior of external tendon from an internal unbonded tendon system. These two features make the stresses in the external tendons member dependent rather than section dependent.

To date, many theoretical and experimental investigations have been undertaken to study the flexural behavior of beams prestressed with unbonded tendons. Some prediction equations have been developed to calculate the tendon stress at ultimate, which have been reviewed and evaluated by many researchers (Naaman and Alkhairi 1991a; Ament et al. 1997; Harajli 2006; Manisekar and Senthil 2006; Dall'Asta et al. 2007). However, there is no such method that gains wide recognition. A noteworthy story is the shift of the related provisions in AASHTO LRFD specifications; that is, the "Naaman method" in the first edition [Naaman and Alkhairi 1991b; American Association of State Highway and Transportation Officials (AASHTO) 1994] was replaced by the "MacGregor method" in the third edition (MacGregor 1989; AASHTO 2004; Roberts-Wollmanm and Kreger 2005). It indicates that the researchers all around the world are trying to develop better methods that have clear physical significance and are practical to designers.

This study attempts to provide a rational solution to this problem. A unified methodology is proposed to calculate stress increment of external and internal unbonded tendons under both service and ultimate limit states, and simple design equations are developed for practical use.

Fundamental Equations

A simply supported concrete beam with constant section, symmetrical loading, and symmetrical profile unbonded tendons is illustrated in Fig. 1(a). The total elongation of the unbonded tendons under applied loads can be taken as the integral of the concrete strains at the level of the tendon. The following assumptions are made in this investigation:

- Shear deformation of the concrete beam is ignored in the elastic range; and
- Stress along the tendon is uniform, friction between the tendon and the duct being neglected.

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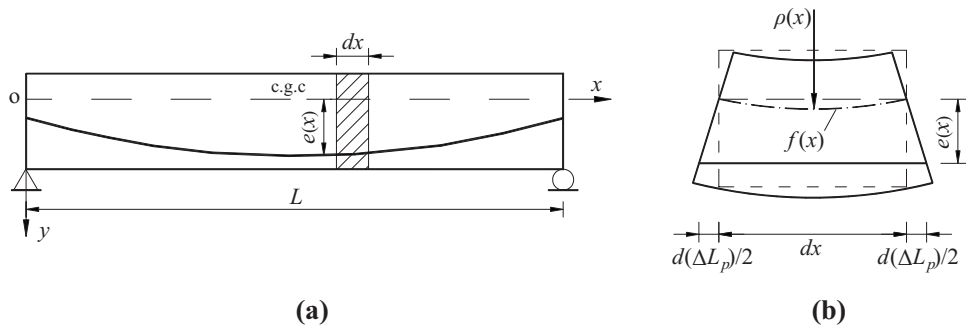

Fig. 1. Simply supported beam prestressed with unbonded tendons: (a) unloaded state; (b) defected shape of beam segment after loading

From Fig. 1(b), tendon elongation along segment dx can be determined by deformation compatibility

$$d(\Delta L_p) = \frac{e(x)}{\rho(x)}dx \approx -e(x)f''(x)dx \tag{1}$$

where ΔL_p=total tendon elongation; $e(x)$=tendon eccentricity along beam span; $f(x)$=beam deflected shape; and $\rho(x)$=radius of curvature.

Thus, the total tendon elongation between end anchorages can be integrated as follows:

$$\Delta L_p = -\int_0^L e(x)f''(x)dx \tag{2}$$

Therefore, tendon stress increment can be computed in the following general manner:

$$\Delta f_{ps} = \frac{E_{ps}\Delta L_p}{L} = -\frac{E_{ps}}{L}\int_0^L e(x)f''(x)dx \tag{3}$$

where E_{ps}=modulus of elasticity of prestressing steel.

Theoretically, Eq. (3) is applicable to different tendon profiles as embodied in the term $e(x)$, as well as to different load types, which is implied in beam deformation curve $f(x)$. Peculiarly, in case of a beam with straight internal unbonded tendons, the tendon eccentricity along the span is constant. Thus Eq. (3) can be simplified as

Straight internal unbonded tendons

$$\Delta f_{ps} = \frac{E_{ps}e_m}{L}(\theta|_{x=0} - \theta|_{x=L}) = \frac{2E_{ps}e_m\theta|_{x=0}}{L} \tag{4}$$

where e_m=tendon eccentricity at beam midspan and $\theta|_{x=0}$ and $\theta|_{x=L}$=angles of rotation of the two supporting sections, respectively.

As shown in Fig. 2, some researchers (MacGregor 1989; Du and Liu 2003) introduced a broken line to simplify the deformation curve of the beam at ultimate. The analytic model consisted of two simply supported rigidly connected bodies linked by a hinge at midspan, which is adopted as the basis for the method in AASHTO (2004) (Roberts-Wollmanm and Kreger 2005). Based on this assumption, the rotation angles of sections at two supports equal to $2\delta_{mid}/L$ (δ_{mid}=beam midspan deflection); thus the tendon stress increment can be derived from Eq. (4) as follows:

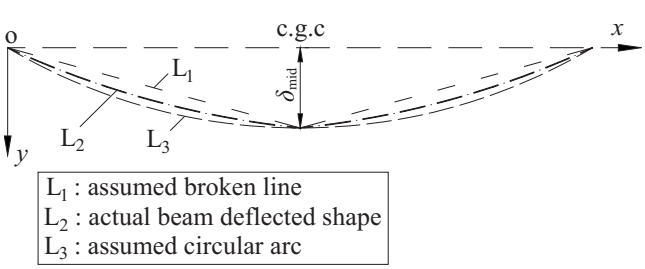

Fig. 2. Assumed and actual beam defected shapes

$$\Delta f_{ps} = \frac{4E_{ps}e_m\delta_{mid}}{L^2} \tag{5}$$

Alternatively, some other researchers (Wang et al. 2005) assumed the deformation curve as a circular arc. The rotation angles of the two end sections equal to $4\delta_{mid}/L$, and the tendon stress increment can be given by

$$\Delta f_{ps} = \frac{8E_{ps}e_m\delta_{mid}}{L^2} \tag{6}$$

Nevertheless, the actual beam deformation curve lies between the broken line and the circular arc illustrated in Fig. 2. Taking the format of Eqs. (5) and (6), tendon stress increment can be expressed as the following form:

Straight internal unbonded tendons

$$\Delta f_{ps} = \eta\frac{E_{ps}e_m\delta_{mid}}{L^2} \tag{7}$$

where η=4–8. By making a comparison between Eqs. (3) and (7), the parameter η can be derived as

$$\eta = -\frac{L}{e_m\delta_{mid}}\int_0^L e(x)f''(x)dx \tag{8}$$

Proposed Methodology in Elastic State

Stress Increment of External Tendons

Motivated by the idea of the tendon stress increment may be preferably expressed in term of the beam midspan deflection rather than the magnitude of loading under a certain load type, as shown in Eq. (7). In the following part, efforts are made to establish the relationship between Δf_{ps} and δ_{mid} for some typical tendon profiles and load types based on the fundamental equation [Eq. (3)].

First, the term $f''(x)$ in Eq. (3) can be obtained from the essential equation for flexural members

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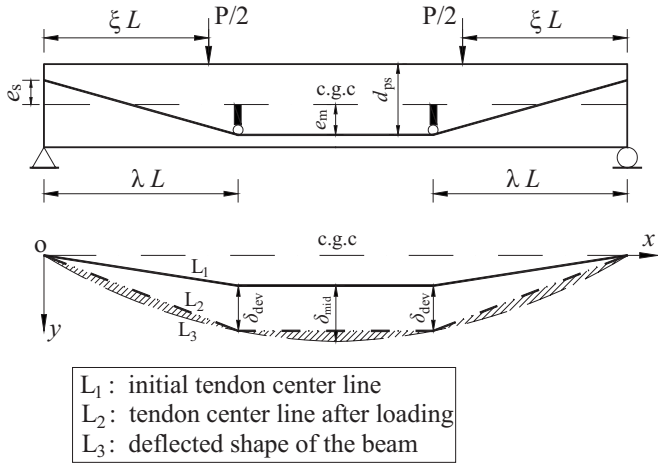

Fig. 3. Schematic deformation curve of concrete beam with harped external tendons

$$f''(x) = \frac{-M(x)}{E_c I_0} \tag{9}$$

where E_c=modulus of elasticity of concrete; I_0=moment of inertia of the beam cross section; and $M(x)$=bending moment along the span due to exterior load. Second, under specific load type, the midspan deflection of linear elastic beam, $\delta_{mid,e}$, can be obtained by the integration of Eq. (9)

$$\delta_{mid,e} = K\frac{M_{\max}L^2}{E_c I_0} \tag{10}$$

where $M_{\max}$=bending moment at midspan section and K=coefficient that relates $\delta_{mid,e}$ and $M_{\max}$; the values of K are assigned to be 1/12, 23/216, and 5/48 under one-point load, two-third-point loads, and uniform load, respectively.

Third, the term $e(x)$ in Eq. (3) should be modified to include the second-order effect in externally prestressed beams. Notice that there is a significant difference between external and internal unbonded tendon systems (Harajli et al. 1999; Ng and Tan 2006a). For beams prestressed with internal unbonded tendons, the tendon eccentricity along the span remains unchanged under loading. On the contrary, the external tendons are free to move relative to the axis of the beam between anchorages and/or deviators, which leads to a progressive change in eccentricity of the tendons as the member deformation increases. This eccentricity variation of external tendons is commonly referred as the second-order effect. Fig. 3 shows a simply supported concrete beam prestressed with harped external tendons. Allowing for the second-order effect, the effective tendon eccentricity can be expressed as

$$e(x) = e_0(x) - \Delta \tag{11}$$

where $e_0(x)$ and $e(x)$=initial and effective tendon eccentricities along beam span, respectively, and Δ=change of eccentricity due to the second-order effect (i.e., the shadow area shown in Fig. 3). Thus, the effective tendon eccentricity can be determined according to the deflections of the adjacent deviators as

$$e(x) = \begin{cases} \dfrac{e_s(x-\lambda L)+e_m x}{\lambda L} - \left[f(x) - \dfrac{\delta_{dev}x}{\lambda L}\right], & x \in [0,\lambda L] \\ e_m - [f(x) - \delta_{dev}], & x \in [\lambda L, L/2] \end{cases} \tag{12}$$

where e_m and e_s=tendon eccentricities at support and midspan of the beam, respectively (e_m and e_s are positive when tendons are below the center of gravity of concrete (c.g.c.) line at midspan and above that at support, respectively); L=effective span of the beam; $\lambda = L_d/L$; L_d=distance from the beam support to the deviation point; and δ_{dev}=beam deflection at the deviation point.

Accordingly, the tendon stress increment can be figured out through the following procedure. For a given load type, the moment $M(x)$ due to this load can be determined. Then, the beam deflection $f(x)$ can be obtained by the integration of Eq. (9) and can be rewritten in the form containing the term $\delta_{mid,e}$ (Balaguru, 1981). Let $\delta_{dev}=f(\lambda L)$ as shown in Fig. 3. Finally, substituting Eqs. (9) and (12) into Eq. (3), the stress increment of the external tendons can be derived as follows:

$$\Delta f_{ps} = \frac{E_{ps}\delta_{mid,e}(\eta e_m - \phi\delta_{mid,e})}{L^2} \tag{13}$$

where η and ϕ can be given for three most common load cases:

1. For one-point load

$$\eta = 6 - 8(1+\beta)\lambda^2 \tag{14a}$$

$$\phi = 4.8 - 18\lambda + 48\lambda^3 - 32\lambda^5 \tag{14b}$$

where $\beta = e_m/e_s$, which is the ratio of the end eccentricity to the midspan eccentricity of the tendons.

2. For two-third-point loads

$$\eta = \begin{cases} 6.26[1 - 3(1+\beta)\lambda^2/2], & \lambda \in [0,1/3] \\ 6.26\{1 - (1+\beta)[1/(9\lambda) - 1 + 3\lambda]/2\}, & \lambda \in [1/3,1/2] \end{cases} \tag{14c}$$

$$\phi = \begin{cases} 4.95 - 19.6\lambda + 58.8\lambda^3 - 44.1\lambda^5, & \lambda \in [0,1/3] \\ -0.06/\lambda + 8.22 - 47.4\lambda + 88.2\lambda^2 - 44.1\lambda^3, & \lambda \in [1/3,1/2] \end{cases} \tag{14d}$$

3. For uniform load

$$\eta = 6.4[1 - (1+\beta)(2\lambda^2 - \lambda^3)] \tag{14e}$$

$$\phi = 4.97 + 20.5(-\lambda + 4\lambda^3 - 2\lambda^4 - 4\lambda^5 + 4\lambda^6 - \lambda^7) \tag{14f}$$

It is noted that for beam with internal unbonded tendons, the parameter ϕ in Eq. (13) equals 0. From Eq. (14), it can be observed that the parameters η and ϕ depend on the load type, the tendon profile, and the ratio of end eccentricity to maximum eccentricity. Accordingly, Eq. (13) is a unified equation applicable to beams with external and internal unbonded tendons, with different tendon profiles and under different load types.

Effective Eccentricity of External Tendons

By now, the second-order effect is considered in the calculation of tendon stress. In addition, the tendon eccentricity variation due to this secondary effect shall also be included in calculating nominal flexural resistance of the beam. The relationship between the loss

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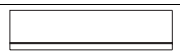
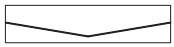
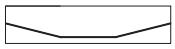
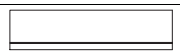
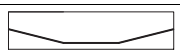
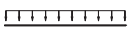
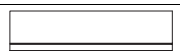
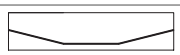
Load case	Tendon profile	η	ϕ	k_1
		6	4.80	1
		$4-2\beta$	0.80	0
		$\frac{46-8\beta}{9}$	0.45	0.148
		$\frac{144}{23}$	4.95	1
		$\frac{88-56\beta}{23}$	0.94	0
		$\frac{100-24\beta}{23}$	0.41	0.130
		$\frac{32}{5}$	4.97	1
		$4-2.4\beta$	0.97	0
		$\frac{32(22-5\beta)}{135}$	0.43	0.131

Fig. 4. Values of η, ϕ, and k_1 under some typical load types and tendon profiles

of tendon eccentricity at beam midspan section Δ_{mid} and beam midspan deflection δ_{mid} can be established according to Eq. (12)

$$\Delta_{mid} = \delta_{mid} - \delta_{dev} = k_1 \delta_{mid} \tag{15}$$

where k_1 is given for three most common load cases:

1. For one-point load

$$k_1 = 1 - 3\lambda + 4\lambda^3 \tag{16a}$$

2. For two-point loads

$$k_1 = \begin{cases} \dfrac{3\xi - 4\xi^3 - 12\lambda\xi + 12\lambda\xi^2 + 4\lambda^3}{\xi(3-4\xi^2)}, & \xi \geqslant \lambda \\ \dfrac{3(1-4\lambda+4\lambda^2)}{3-4\xi^2}, & \xi \leqslant \lambda \end{cases} \tag{16b}$$

3. For uniform load

$$k_1 = 1 - 16\lambda(1 - 2\lambda^2 + \lambda^3)/5 \tag{16c}$$

Parametric Evaluation of η, ϕ, and k_1

From Eqs. (13) and (15), it is interesting to note that the behavior of externally prestressed beams could be well characterized by such three geometry-dependent parameters as η, ϕ, and k_1. That is, the parameter η reflects the relationship between the tendon stress increment and the beam midspan deflection; the parameter ϕ represents the influence of the second-order effect to the tendon stress increment; and the parameter k_1 expresses the loss of tendon eccentricity at beam midspan section due to the second-order effect. Applying Eqs. (14) and (16) to some typical load types and tendon profiles leads to the values of η, ϕ, and k_1 listed in Fig. 4. Parametric evaluation of η, ϕ, and k_1 for different load types and tendon profiles is also shown in Fig. 5. It can be observed that the load type has no significant influence on the values of these parameters.

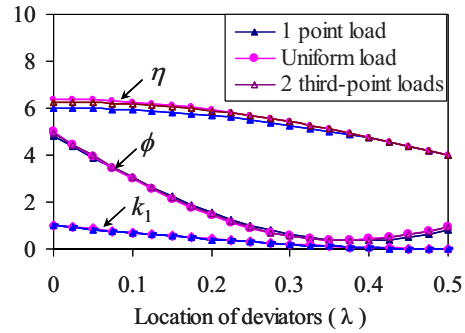


Fig. 5. Parametric evaluation of η, ϕ, and k_1 (assuming $\beta=0$)

Relationship between Δf_{ps} and $\delta_{mid,e}$

So far, the relationship between the tendon stress increment and the beam midspan deflection has been established in Eq. (13). For internal unbonded tendons, $\phi=0$, and Eq. (13) reduces to the following linear equation:

internal unbonded

$$\Delta f_{ps} = \frac{\eta E_{ps} e_m}{L^2} \cdot \delta_{mid,e} \tag{17}$$

For external tendons, Eq. (13) is a quadratic equation for the consideration of the second-order effect. However, the value of the term $\phi\delta_{mid,e}$ is much smaller than that of ηe_m in the case of elastic analysis or if deviators are set well along the beam span. Based on the above information, it may be practically concluded that there is an almost linear relationship between the tendon stress increment and beam midspan deflection in the elastic range.

Proposed Methodology at Ultimate Limit State

Experimental Observation

In the linear elastic range of beams with unbonded tendons, it has been concluded that there is an almost linear relationship between Δf_{ps} and δ_{mid}. Moreover, many tests have shown that this linear relationship is valid throughout the entire loading phase even if the beam is near the stage of collapse. Experimental studies conducted by Du and Tao (1985) showed that there exists such a relationship for moderately reinforced partially prestressed concrete beams until damage. Experimental tests by Campbell and Chouinard (1991) and Du and Liu (2003) confirmed this observation. Based on statistical analysis, Alkhairi (1991) concluded that beams with reinforcing indices ranging from 1.5 to 0.30% showed a linear Δf_{ps} versus δ_{mid} response. Yang (2005) also found out that the curves of load versus deflection and load versus increase of tendon stress in the externally prestressed beams were very similar in shape.

Based on some test results of Harajli and Kanj (1991), Mutsuyoshi et al. (1995), and Yaginuma (1995), the relationship between experimental Δf_{ps} and δ_{mid} is plotted in Fig. 6 throughout the whole loading stages. As indicated by Fig. 6, the linear correlation between them is highly tenable.

Stress Increment at Ultimate Strength Limit State

Based on the experimental observations mentioned above, Eq. (13) is also valid for calculating the tendon stress increment at ultimate as follows:

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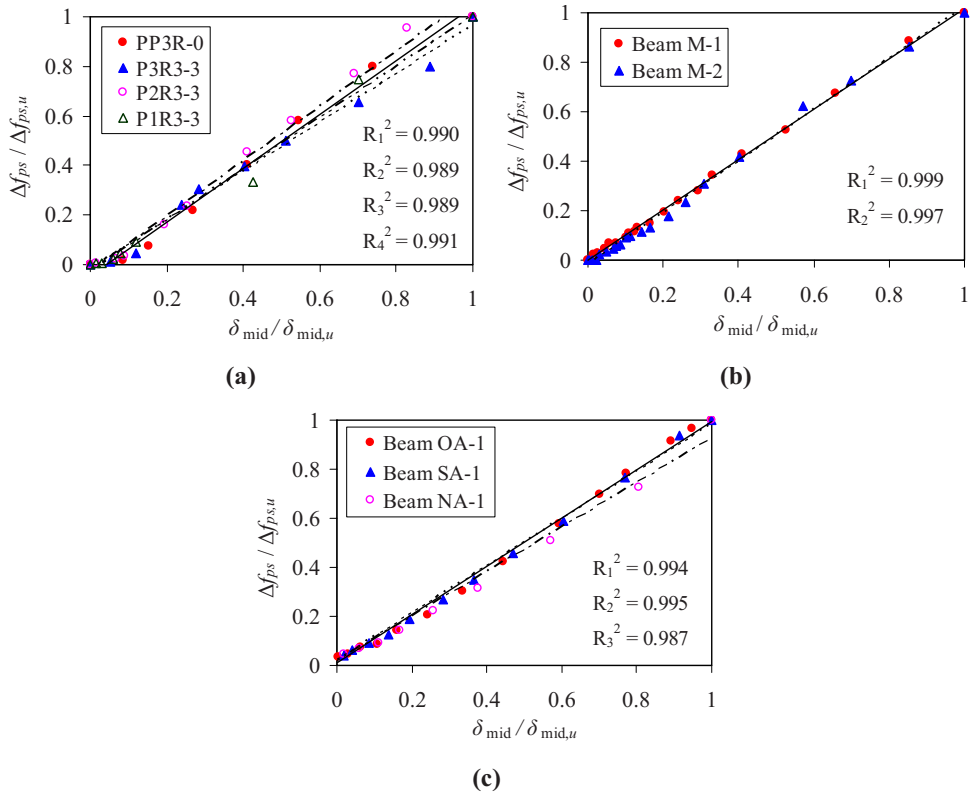

Fig. 6. Experimental Δf_{ps} versus δ_{mid}: (a) beams tested by Harajli and Kanj; (b) by Mutsuyoshi et al.; and (c) by Yaginuma

stress increment at ultimate

$$\Delta f_{ps,u} = \frac{E_{ps}\delta_{mid,u}(\eta e_m - \phi\delta_{mid,u})}{L^2} \tag{18}$$

where $\delta_{mid,u}$=beam midspan deflection at the strength limit state and the parameters η and ϕ are the same as given in Eq. (14). Enlightened by researchers such as Du and Liu (2003), Ghallab and Beeby (2004), and Ng and Tan (2006b), beam deflection $\delta_{mid,u}$ can be given as the following equation that takes the similar format as Eq. (10):

$$\delta_{mid,u} = \gamma\frac{KM_uL^2}{E_cI_e} = \gamma KL^2\phi_u = \gamma KL^2\frac{\varepsilon_{cu}}{c} \tag{19}$$

where γ=deflection reduction factor, further discussion on this factor will be carried out in the next section; I_e=effective moment of inertia of the midspan section at ultimate; M_u=ultimate moment of resistance; ϕ_u=curvature at the midspan section at ultimate; ε_{cu}=strain in the top concrete fiber at ultimate, which is usually taken as 0.003; and c=neutral axis depth.

Substituting Eq. (19) into Eq. (18), an equation for calculating the ultimate stress increment of external and internal unbonded tendons can be expressed as

$$\Delta f_{ps,u} = \gamma E_{ps}\varepsilon_{cu}\eta K\frac{e_m}{c}\cdot R_s \tag{20}$$

As proved in the Appendix of this paper, $\eta K=\Omega_e$. Thus Eq. (20) can be rewritten as

$$\Delta f_{ps,u} = \gamma E_{ps}\varepsilon_{cu}\Omega_e\frac{e_m}{c}\cdot R_s \tag{21}$$

where Ω_e=bond reduction coefficient in the elastic state defined by Naaman and Alkhairi (1991b) and R_s=stress increment reduction factor to consider the second-order effect. For internal unbonded tendons, $R_s=1$. Moreover, for external tendons, R_s is given as

$$R_s = 1 - \frac{\phi\delta_{mid,u}}{\eta e_m} \tag{22}$$

Moreover, from equilibrium condition of the section forces at the ultimate resistance, for a T section, the neutral axis depth c can be given as

$$c = \frac{A_{ps}f_{ps,u} + A_sf_y - A_s'f_y' - 0.85\beta_1f_c'(b-b_w)h_f}{0.85\beta_1f_c'b_w} \tag{23}$$

where b and b_w=beam width and beam web width, respectively; h_f=flange thickness of beam; A_{ps}=area of unbonded tendons; A_s and A_s'=areas of internal tension and compression reinforcement, respectively; f_y and f_y'=yield strengths of internal tension and compression reinforcement, respectively; f_c'=concrete cylinder compressive strength; $f_{ps,u}$=ultimate tendon stress of unbonded tendons, $f_{ps,u}=f_{pe}+\Delta f_{ps,u}$; and β_1=compression stress block depth factor, according to ACI 318 code, $\beta_1=0.85-0.05(f_c'-28)/7\geqslant 0.65$. Finally, tendon stress increment at ultimate can be obtained by a closed-form solution of Eqs. (18), (19), and (23) or Eqs. (19) and (21)–(23); the procedure to facilitate hand calculations is established in the Appendix of this paper.

Determination of the Deflection Reduction Factor

To use Eq. (19) to calculate the ultimate beam midspan deflection $\delta_{mid,u}$, there is an assumption that stiffness of any cross section along the beam span is equal to that of the midspan section, which might lead to an overestimated deflection. This issue is addressed by multiplying a deflection reduction factor γ in Eq. (19). On the other hand, many previous studies (see Table 1) had adopted the

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Table 1. Expressions for L_p and the Corresponding $\delta_{mid,u}$

Source	Expression for L_p	General form of L_p	Resulting expression for $\delta_{mid,u}$
Tam and Pannell 1976	$10.5c$		$\frac{\alpha_1\varepsilon_{cu}}{4}\cdot L$
Au and Du 2004	$16.1c$, $10c$, $9.3c$	$\alpha_1\cdot c$	
Harajli 2006	$\left[\frac{20.7}{f}+10.5\right]c$		
Harajli 1990	$\left[\frac{0.95}{f}+0.05+\frac{1}{L/d_{ps}}\right]L$		$\frac{\alpha_2\varepsilon_{cu}}{4}\cdot\frac{L^2}{c}$
Lee et al. 1999	$\left[\frac{1}{f}+\frac{1}{L/d_{ps}}\right]L$	$\alpha_2\cdot L$	
Harajli et al. 2002	$\left[\frac{1}{f}+\frac{1.5}{L/d_{ps}}\right]L$		
Naaman and Alkhairi 1991b	$1.5d_{ps}$, $3.0d_{ps}$	$\alpha_3\cdot d_{ps}$	$\frac{\alpha_3\varepsilon_{cu}}{4}\cdot\frac{d_{ps}}{c}\cdot L$

plastic hinge length method to calculate the ultimate beam deflection. Based on the model shown in Fig. 7, $\delta_{mid,u}$ can be given as

$$\delta_{mid,u}=\frac{\varepsilon_{cu}L_pL}{4c} \tag{24}$$

where L_p=plastic hinge length, which is difficult to be accurately determined (Manisekar and Senthil 2006; Harajli 2006). From Table 1, it can be seen that these equations for L_p can be classified into three categories: the plastic hinge length L_p is proportional to either neutral axis depth c, beam length L, or tendon depth d_{ps}. Accordingly, the resulting $\delta_{mid,u}$ is proportional to either L, L^2/c, or $d_{ps}L/c$.

Considering that the method proposed by Lee et al. (1999) yields $\delta_{mid,u}$ that has the same format with Eq. (19), it is adopted to determine the deflection reduction factor herein. The plastic hinge length ratio is expressed as

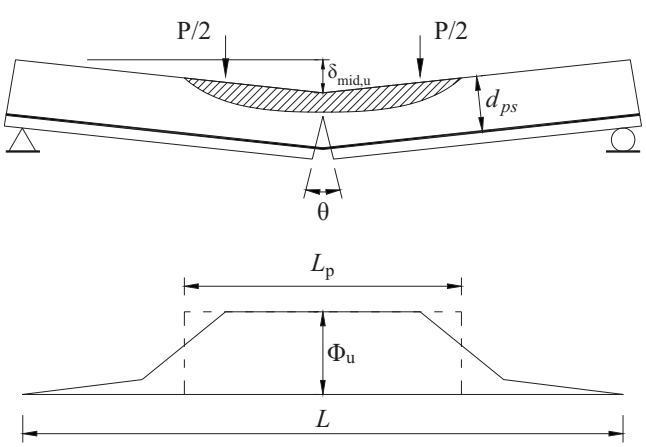

Fig. 7. Failure mechanism and idealized curvature distribution

$$\frac{L_p}{L}=\frac{1}{f}+\frac{1}{L/d_{ps}} \tag{25}$$

where f=10 for one-point load and f=3 for two-point loads or uniform load.

Comparing Eqs. (24) and (25) with Eq. (19), the deflection reduction coefficient γ can be obtained as

$$\gamma=\frac{L_p/L_0}{4K}=\frac{1}{4K}\left(\frac{1}{f}+\frac{1}{L/d_{ps}}\right) \tag{26}$$

where $\gamma=0.3+3/(L/d_{ps})$ for one-point load and $\gamma=0.78+2.35/(L/d_{ps})$ for two-third-point loads or uniform load.

Practical Design Equations

Design Equation for Ultimate Stress

The methodology proposed in previous section is able to calculate $\Delta f_{ps,u}$ accurately. However, simplified prediction equation may be meaningful for practical purposes. The following simplification and improvement are made to Eq. (21):

1. The parameter Ω_e in Eq. (21) depends on the load type, the tendon profile, and the ratio of end eccentricity to maximum eccentricity of the beam (Naaman and Alkhairi 1991b). Although this parameter has been well established, it is still tedious for practical use. Based on the common practice, the beam with external tendons harped at third point ($\lambda=1/3$) or with internal unbonded tendon having a parabolic profile and with tendons anchored on the c.g.c line at the beam end ($\beta=0$) is selected as the representative beam. With this simplification, the bond reduction coefficient Ω_e's for third-point load (or uniform load) and one-point load are approximately 0.54 and 0.42, respectively.
2. According to the recommendation of ACI-ASCE Committee 423 (Naaman et al. 2002), the ratio L_1/L_2 is introduced to consider the influence of loaded span or spans (L_1) versus whole spans (L_2) in continuous beams.

Thus, a design equation for calculating the ultimate stress of unbonded prestressing steel or fiber-reinforced plastic (FRP) tendons is proposed as follows:

$$f_{ps,u}=f_{pe}+\Delta f_{ps,u}\leqslant 0.8f_{pu} \tag{27a}$$

$$\Delta f_{ps,u}=\kappa E_{ps}\frac{e_m}{c}\cdot R_s\cdot\frac{L_1}{L_2} \tag{27b}$$

where $f_{ps,u}$=stress in unbonded tendons at nominal bending resistance; f_{pe}=effective prestress in the prestressed reinforcement; $\Delta f_{ps,u}$=stress increment in unbonded tendons at nominal bending resistance; f_{pu}=specified tensile strength of prestressed tendons; E_{ps}=elastic modulus of prestressing steel or FRP; $\kappa=1.62\times10^{-3}[0.78+2.35/(L/d_{ps})]$ for uniform or two-third-point loads and $\kappa=1.26\times10^{-3}[0.3+3/(L/d_{ps})]$ for one-point midspan load; c=neutral axis depth (for simplification, it can be calculated assuming all mild and prestressed reinforcing steel is at yield (Roberts-Wollmanm and Kreger 2005): $c\approx c_y$, where c_y can be obtained by replacing $f_{ps,u}$ in Eq. (23) with $0.8f_{pu}$); R_s=stress increment reduction factor for considering the second-order effect in external tendons (for internal unbonded tendons, $R_s=1$); and L_1/L_2=ratio of the length of loaded span or spans (L_1) to the total length of tendon (L_2) between anchorages in continuous beams.

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Reduction Factors for External Tendons

The term $\delta_{mid,u}$ in Eq. (22) related to the stress increment reduction factor R_s can be determined by a closed-form solution. However, it is time consuming for practical application. As shown in Table 1, equation of $L_p=\psi c$ by Tam and Pannell leads to $\delta_{mid,u}$ that has a simple format just proportional to the beam length L. Au and Du (2004) further recommended the use of a value of $\psi=16.1$ for relatively better correlation between the calculated and experimental results. Thus, $\delta_{mid,u}$ equals to $0.012L$, and Eq. (22) can be simplified as

$$R_s = 1 - \frac{L/e_m}{83.3}\frac{\phi}{\eta} \tag{28}$$

Parametric evaluation shows that load type has no significant influence on the values of η, ϕ, and k_1, thus η and ϕ in the above equation can be taken as that of the one-point load: $\eta=6-8\lambda^2(1+\beta)$ and $\phi=4.8-32\lambda^5+48\lambda^3-18\lambda$. A depth reduction factor R_d is also introduced to consider the change of the tendon depth due to the second-order effect at ultimate. According to Eq. (15) and the above simplification to $\delta_{mid,u}$, R_d can be expressed as

$$d_{ps,u} = R_d d_{ps} \tag{29}$$

$$R_d = 1 - 0.012k_1\frac{L}{d_{ps}} \tag{30}$$

where d_{ps} and $d_{ps,u}$=initial tendon depth and effective tendon depth at ultimate, respectively, and k_1 can be taken as that of the one-point load: $k_1=4\lambda^3-3\lambda+1$.

Experimental Verification

Internal Unbonded Tendons

To verify the proposed methods and make a comparison to the American codes [AASHTO 1994, 2004; American Concrete Institute (ACI) 2008], a set of data including 89 test beams prestressed with internal unbonded tendons (Tam and Pannell 1976; Du and Tao 1985; Harajli and Kanj 1991; Chakrabarti 1995) is gathered together and plotted in Fig. 8. The accuracy of the predictions is evaluated in terms of mean value and standard deviation. From Fig. 8(a), the proposed methodology is found to give relatively accurate results: the experimental ultimate stress increment is on average 134% of the predicted value, with a standard deviation of 46%. Moreover, Fig. 8(b) shows that the proposed simplified equation does not lose much accuracy compared to the proposed methodology. The AASHTO (1994, 2004) specifications are both very conservative compared to the experimental results. The high standard deviations show a scattering phenomenon. However, the ACI 318 code is shown to be unconservative in some cases. Based on the experimental evaluation, it is clear that the proposed equations are able to predict $\Delta f_{ps,u}$ of internal unbonded tendons more consistently and accurately, which is evident in the relatively low standard deviation and the better correlation between the experimental data and predicted results.

External Tendons

Nine externally prestressed beams tested by Ng and Tan (2006b) are selected to verify the theoretical methods. The deviators in these beams are arranged as no deviator was provided in Beams T-0A and T-0B; a single deviator located at midspan was provided in Beams ST-1–ST-5; two deviators located at third points were provided in Beam ST-5A; and three deviators located at quarter points are set in Beam ST-5B. Fig. 9 shows the comparison between the test results and the predictions of different equations. Both the proposed methodology and design equation are shown to have a better agreement with the test results compared to the American codes.

Evaluation of Second-Order Effects

Evaluation of Stress Reduction Factor R_s

The stress reduction factor R_s in Eq. (28) characterizes the influence of the second-order effect to the ultimate tendon stress increment. Fig. 10 illustrates the influence of some critical variables to this secondary effect. The following conclusions can be drawn based on Eq. (28) and Fig. 10:

1. The magnitude of the second-order effect is mainly dependent on the arrangement of deviators along the span (λ), the profile of the tendons (β), and the span-to-eccentricity ratio (L/e_m).
2. If no deviator is placed, the reduction in tendon stress increment due to the second-order effect can be as high as 50% for beams with span-to-eccentricity ratios of 50 or higher.
3. The second-order effect can be effectively reduced as long as a single deviator is placed at the beam midspan.
4. If two deviators are placed near the third point along the span, this secondary effect will be less than 5% for beams with span-to-eccentricity ratios of up to 50.
5. Optimization of the arrangement of the deviators can be taken to minimize the second-order effect so that the beams would achieve the desired flexural performance. For beams with up to two deviators placed symmetrically about the midspan of the beam, the optimal location of these deviators can be obtained using $\partial R_s/\partial\lambda=0$. The optimal values of λ for some typical load types and tendon profiles are listed in Table 2.

The above conclusions are consistent with some previous numerical and experimental studies (Tan and Ng 1997; Harajli et al. 1999; Wu and Lu 2003; Lou and Xiang 2006). In order to verify the above conclusions further, a comparative study is conducted based on the nine beams tested by Ng and Tan (2006b). It can be seen from Fig. 11 that, for Beams ST1–ST5, ST-5A, and ST-5B with at least one deviator, the second-order effect can be neglected in practice. For Beams T-0A and T-0B with no deviator, the predictions would be highly overestimated if not accounting for the second-order effect.

Evaluation of Tendon Depth Reduction Factor R_d

The depth reduction factor R_d in Eq. (30) reflects the influence of the second-order effect to the effective tendon depth at beam midspan section. It can be seen from Eq. (30) that the critical factors include the arrangement of deviators (λ) and the span-to-depth ratio (L/d_{ps}). Comparison is also made with the depth reduction factor proposed by Mutsuyoshi et al. (1995). R_d proposed by them for two-point loading can be written as

$$R_d = 1 - 0.022\cdot(L/d_{ps} - 5)\cdot(0.8 - 2\lambda) \tag{31}$$

Fig. 12 plots the two critical factors (i.e., λ and L/d_{ps}) against R_d for Eqs. (30) and (31). The following conclusions can be made:

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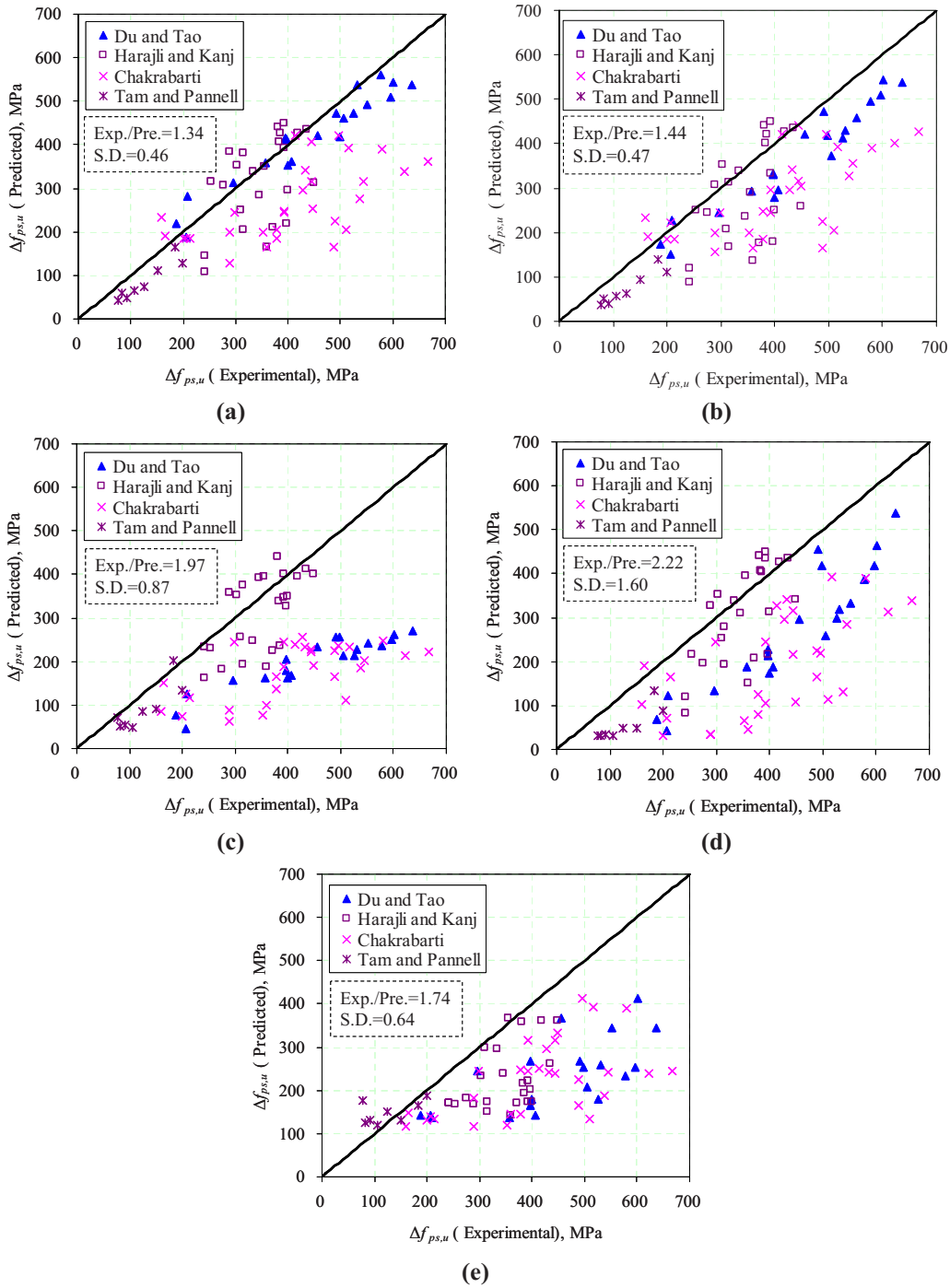

Fig. 8. Comparison of predicted $\Delta f_{ps,u}$ with experimental results: (a) proposed methodology: Eq. (21); (b) proposed prediction equation: Eq. (27); (c) AASHTO LRFD 2004; (d) AASHTO LRFD 1994; and (e) ACI 318-08

1. R_d's predicted by the writers and Mutsuyoshi et al. have a similar value in general. However, proposed Eq. (30) is more reasonable to deal with the situation when one deviator is placed at beam midspan (when $\lambda=0.5$, $R_d=1$).
2. Similarly to R_s, for beams with span-to-depth ratios of higher than 25, if no deviator is provided, the loss of tendon depth due to the second-order effect can be as high as 30%. However, if deviators are set, this secondary effect can be effectively reduced.

Characteristics of Proposed Methodology

The proposed methodology and design equations have many characteristics and advantages as follows:

- First and foremost, the proposed prediction equation for Δf_{ps} is rational and based on a fundamental analysis where equilibrium equation, constitutive equation, and deformation compatibility equation are all satisfied.
- Second, the proposed methodology is characterized in generality and unification: it is applicable to both elastic and ulti-

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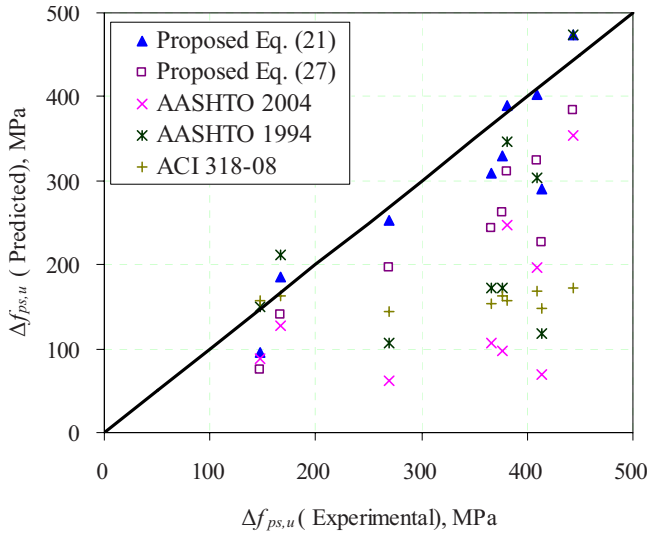

Fig. 9. Comparison of theoretical predictions with test results of Ng and Tan (2006b)

mate strength limit state analyses, both external and internal unbonded tendons, both simply supported and continuous beams, and both prestressing steel and FRP tendons.

- Third, the proposed methodology for stress increment calculation at service and ultimate strength limit states has the same form, as shown in Eqs. (13) and (17). Moreover, it is essentially identical to the Naaman method in the elastic analysis, which is proved in the Appendix of this paper.
- Fourth, the proposed design equation includes almost all the influential factors summarized by Ghallab and Beeby (2005). The ultimate stress increment $\Delta f_{ps,u}$ is mainly dependent on e_m/c. The parameter e_m reflects the shape of the beam section and the tendon eccentricity and the parameter c reflects the rotation capacity of the section and the deformability of the full member. This is one of the most important points found in this study.
- Fifth, the approach accounts for the second-order effect through two reduction factors: the stress reduction factor R_s and the depth reduction factor R_d.
- Finally, the proposed design equation [Eq. (27)] is simple and rational and has a better accuracy than the current American codes.

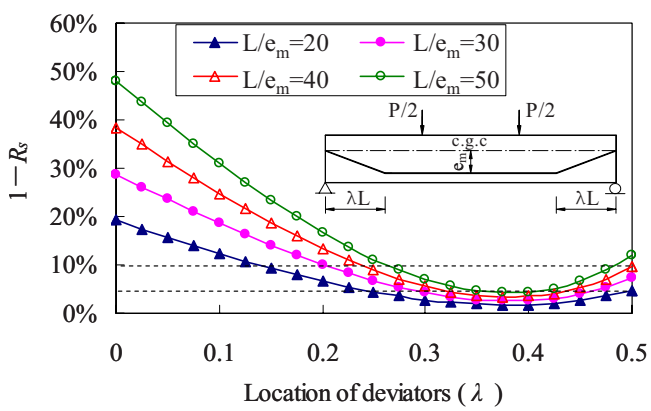

Fig. 10. Effect of deviators arrangement to R_s (assuming $\beta=0$)

Table 2. Optimal Location of Deviators

	$\beta=0$	$\beta=-1$
One-point load	0.382	0.387
Two-third-point loads	0.367	0.379
Uniform load	0.360	0.367

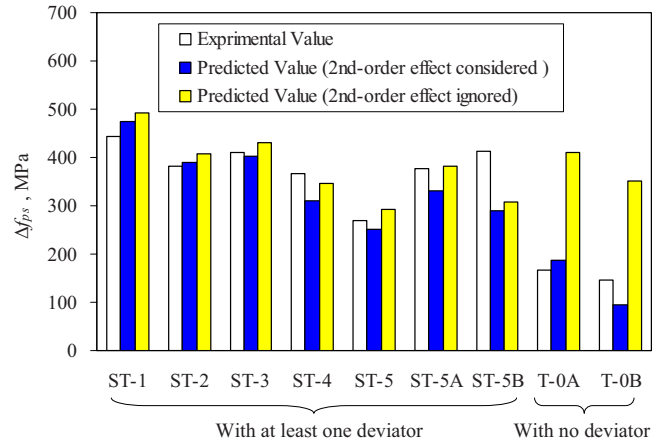

Fig. 11. Comparative analysis of the second-order effect

Summary and Conclusions

A unified methodology is developed to predict stresses in external and internal unbonded tendons under both service and strength ultimate limit states. Practical design equations are put forward to predict the tendon stresses at ultimate. The following conclusions can be made:

1. Previous test results have shown that there is an almost linear relationship between the tendon stress increment and the beam midspan deflection throughout the entire loading stages. This relationship is well established for externally prestressed beams in this study which leads the proposed methodology to be characterized in generality and unification.

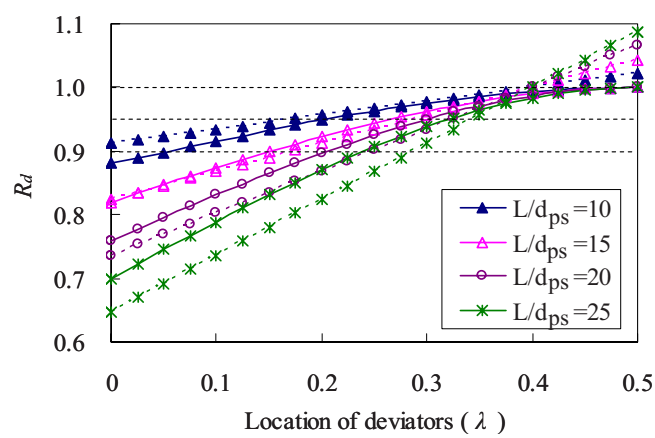

Fig. 12. Effect of deviators arrangement to R_d [solid line: Eq. (30) proposed in this paper; dashed line: Eq. (31) proposed by Mutsuyoshi et al.]

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2. The term Ω_e in Eq. (21) indicates that the proposed methodology is identical to the Naaman method in the elastic analysis.
3. The proposed design equation [Eq. (27)] is able to account for most of the variables considered to be important so far. Moreover, the ultimate stress increment $\Delta f_{ps,u}$ is mainly dependent on e_m/c.
4. The second-order effect in externally prestressed beams can be reasonably considered through two reduction factors: the stress reduction factor R_s and the depth reduction factor R_d. Moreover, the evaluation shows that, if two deviators are placed symmetrically near the third point along the beam span, the second-order effect can be neglected in practice.
5. Eq. (27) is recommended for the calculation of the ultimate stress in prestressing steel and FRP unbonded tendons.

Acknowledgments

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Appendix

Derivation of $\Omega_e=\eta K$

Naaman and Alkhairi (1991b) proposed a unified approach (the Naaman method) to calculate the stress in unbonded tendons in the elastic regime as well as the ultimate resistance. The approach introduced a bond reduction coefficient to reduce the analysis of beam prestressed with unbonded tendons to a conventional sectional analysis. The bond reduction coefficient in the elastic range is expressed as

$$\Omega_e = \frac{(\Delta\varepsilon_{psu})_{avg}}{(\Delta\varepsilon_{cps})_m} \tag{32}$$

where $(\Delta\varepsilon_{psu})_{avg}$=strain increase in the unbonded tendons and $(\Delta\varepsilon_{cps})_m$=strain increase in the concrete at the level of the tendon beyond effective prestress at the critical section of maximum moment.

For simply supported beams with constant section, symmetrical loading, and symmetrical profile, Ω_e can be calculated from

$$\Omega_e = \frac{2}{M_{\max} e_m L}\int_0^{L/2} M(x)e(x)dx \tag{33}$$

Combining Eqs. (8)–(10) and (33) leads to

$$\Omega_e = \eta K \tag{34}$$

The above equation shows that Ω_e can be obtained through the product of η and K that have been defined in this paper. This indicates that the methodology proposed in this paper is essentially the same with the Naaman method in the elastic analysis.

Procedure for Accurately Calculating $\Delta f_{ps,u}$

A closed-form solution is needed to calculate $\Delta f_{ps,u}$ accurately using the proposed methodology. The procedure to facilitate hand calculations is developed herein as follows. The combined reinforcing index $\bar{w}$ to account for the presence of bonded reinforcement and unbonded tendons is defined as follows:

$$\bar{w} = \frac{A_{ps}f_{pe} + A_s f_y - A_s' f_y' - 0.85\beta_1 f_c'(b-b_w)h_f}{0.85\beta_1 f_c' b_w d_{ps}} \tag{35}$$

where the index $\bar{w}$ reflects the neutral axis depth, on which the tendon stress increment at failure depends strongly.

Substitution of Eq. (35) into Eq. (23) yields

$$c = k\bar{w}d_{ps} \tag{36}$$

If notations $\Gamma_1=\gamma\eta K\varepsilon_{cu}E_p e_m/\bar{w}d_{ps}$, $\Gamma_2=\gamma\phi K\varepsilon_{cu}L^2/\eta\bar{w}d_{ps}e_m$, and $\Gamma_3=A_{ps}/0.85\beta_1\bar{w}f_c' b_w d_{ps}$ are defined, Eq. (18) can be rewritten as

$$\Delta f_{ps,u} = \frac{\Gamma_1}{k}\left(1-\frac{\Gamma_2}{k}\right) \tag{37}$$

Combining Eqs. (36) and (37), the value of k can be determined by solving the following cubic equation:

$$k^3 - k^2 - \Gamma_1\Gamma_3 k + \Gamma_1\Gamma_2\Gamma_3 = 0 \tag{38}$$

For beam with internal unbonded tendons, $\Gamma_2=0$, thus $k=0.5(1+\sqrt{1+4\Gamma_1\Gamma_3})$. Therefore, the procedure for stress increment calculation is as follows:

Step 1: Determine the value of η, ϕ, γ, and K according to load type and tendon profile of the beam;

Step 2: Calculate Γ_1, Γ_2 (for beam with internal unbonded tendons, $\Gamma_2=0$), and Γ_3, then solve Eq. (38) to get k; and

Step 3: Calculate $\Delta f_{ps,u}$ using Eq. (37).

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